UPSKILL CAMPUS (USC) & UCT INTERNSHIP

Industrial Internship Report on

”Smart City AI Traffic Pattern Detection”

Prepared by

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UCT Internship -2026

Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time.

My project focused on engineering a production-grade Smart City AI Traffic Hub & Pattern Detection Engine. The system utilizes advanced machine learning to model temporal urban traffic flow, identify rush-hour congestion peaks, and display real-time sequences via a mobile-responsive interface. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solutions for that. It was an overall great experience to have this internship.

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Preface

This report details a 6-week technical program mapping data cleaning, feature engineering, and model validation stages. Strong industrial internships bridge academic learning and real-world application, which is crucial for modern engineering careers. My project tackles urban infrastructure congestion by building an automated, zero-input predictive engine.

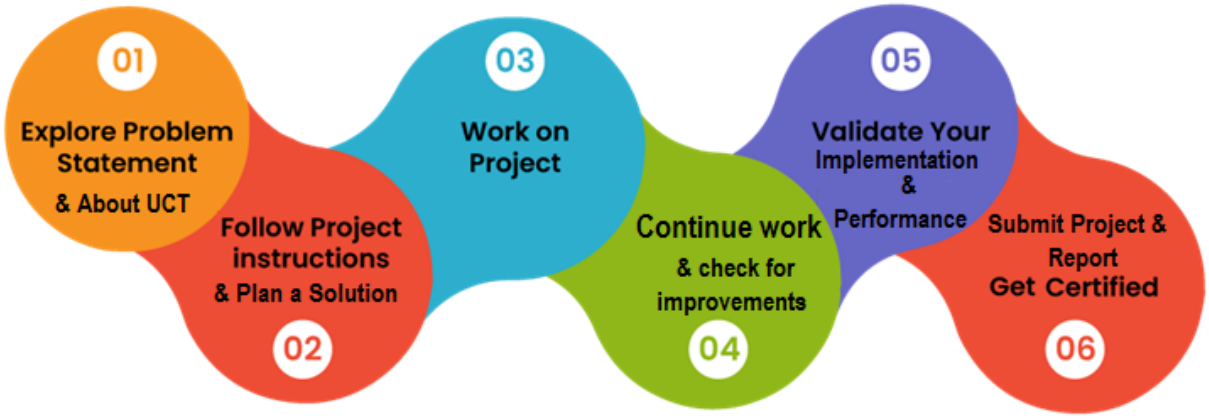
The program was planned sequentially:

• Weeks 1-2: Explore problem statement, gather smart city grid parameters, and study UCT platform systems.

• Weeks 3-4: Clean datasets, perform visual exploratory data analysis, and configure model features.

• Weeks 5-6: Tune gradient boosting pipelines, evaluate unseen validation split matrices, and deploy the application.

I want to extend my deep gratitude to the mentors at Upskill Campus, The IoT Academy, and UniConverge Technologies for their constant structural feedback, as well as my peers who collaborated during testing rounds.



1 Introduction

1.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



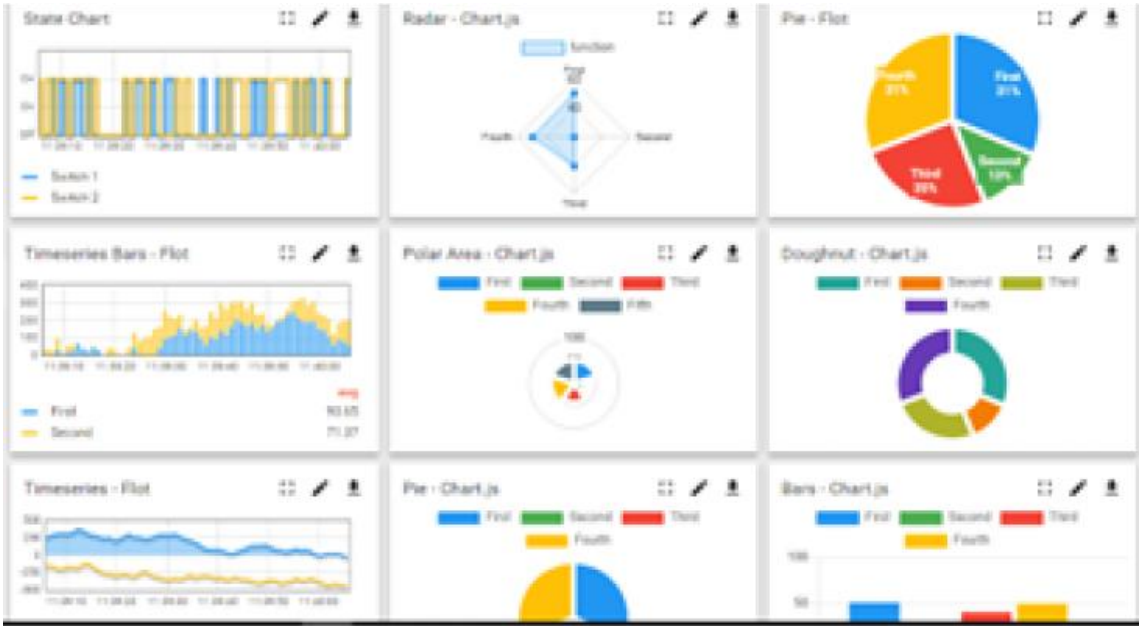
i. UCT IoT Platform (UCT Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

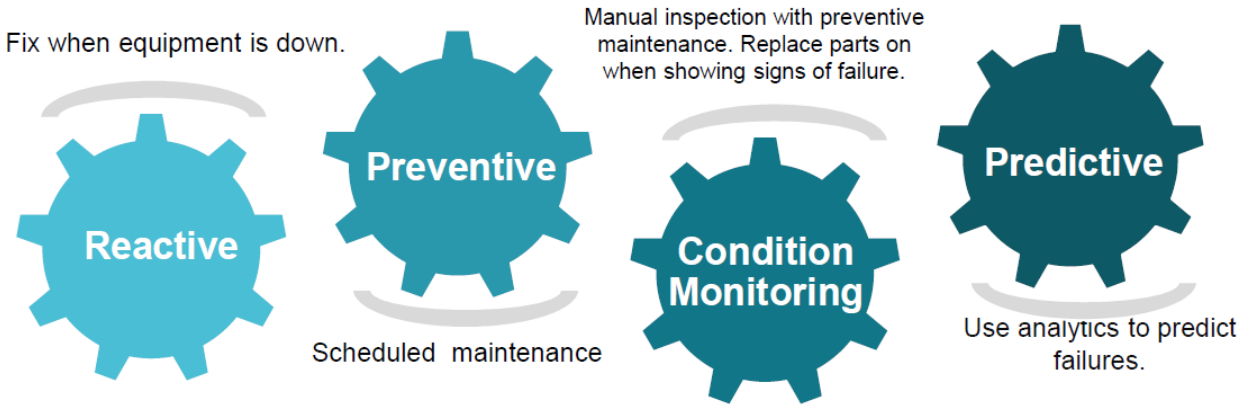


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agri tech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

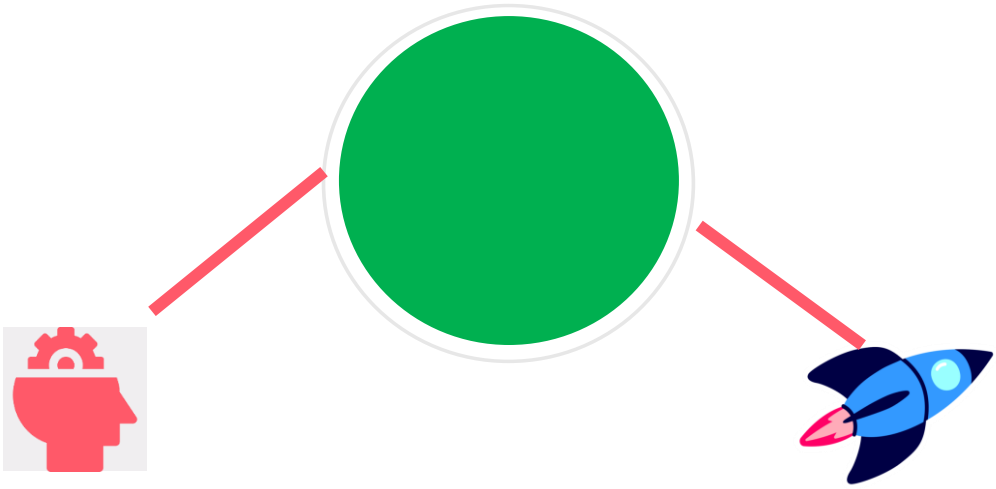
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



1.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

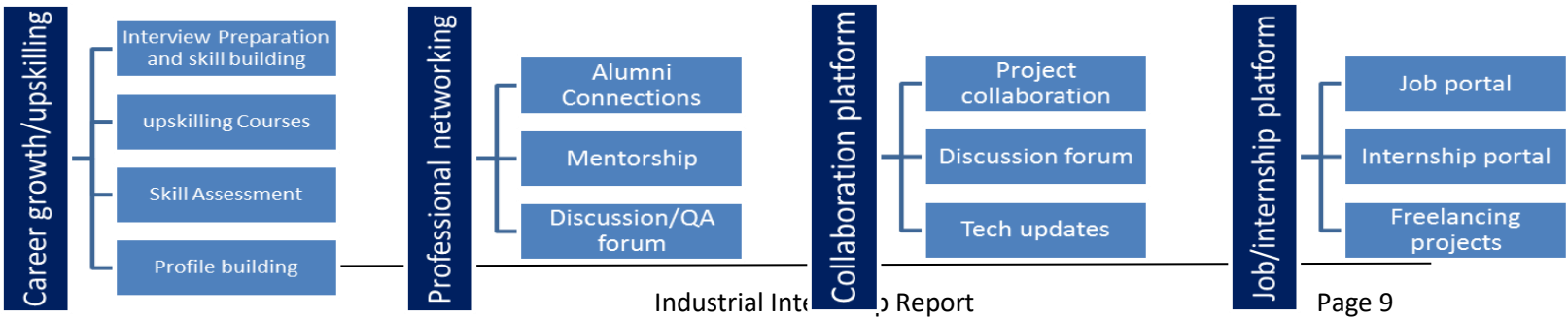
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

Reference

[1] Extreme Gradient Boosting (XGBoost) Framework Documentation.

[2] Streamlit Cloud Micro-Service Infrastructure: Streamlit Open-Source Developer Ecosystem (2026). Application Architecture Framework for low-latency data rendering and UI dashboard compilation.

[3] Cyclical Spatial-Temporal Feature Engineering Standards: Trigonometric encoding methodologies for mapping seasonal data variants without temporal feature compression or date-boundary leakage.

Glossary

Terms	Acronym
Intelligent Transportation Systems (ITS)	Advanced applications are designed to provide innovative services relating to different modes of transport and traffic management.
Exploratory Data Analysis EDA	An approach to analyzing datasets to summarize their main statistical characteristics, often with visual methods
Root Mean Squared Error (RMSE).	A standard statistical metric used to measure the model error rate, emphasizing larger errors or variances
Spatial-Temporal Wave Geometry	The mathematical conversion of raw time/location dimensions into multi-dimensional coordinates to track physical moving distributions over time.
Mean Absolute Error (MAE)	A metric that measures the average magnitude of absolute errors in a set of predictions, without considering their direction.
Extreme Gradient Boosting XGBoost	An optimized distributed gradient boosting library is designed to be highly efficient, flexible, and portable.
Gradient Boosted Decision Tree	The underlying machine learning technique that produces a prediction model in the form of an ensemble of weak decision trees.
Key Performance Indicator (KPI)	Quantifiable measures used to evaluate the success of an operational system (e.g., volumetric aggregates, flow rates).

Problem Statement

Rapid urbanization demands highly responsive Intelligent Transportation Systems (ITS) to minimize municipal traffic congestion, economic transit delays, and excessive vehicular emissions. Traditional corporate traffic architectures fall short because they rely heavily on static historical averages, necessitate manual operators to log data, or inadvertently introduce temporal data leakage during validation, causing high failure rates when deployed live.

Furthermore, urban traffic behaviors are highly non-linear, exhibiting prominent bimodal rush-hour spikes (9:00 AM and 6:00 PM) that standard tracking systems fail to smooth or accurately project across sharp temporal transitions like midnight boundaries. There is an immediate business need for an automated, zero-input predictive engine that utilizes advanced cyclical wave geometry to deliver real-time, 24-hour sequence forecasts across city grid junctions with high accuracy and minimal system latency.

Existing and Proposed solution

• Existing Models: Basic historical metrics use rolling statistics or naive linear regressions. These tools struggle to adapt to cyclic weekly shifts and fail when long-term sequence parameters change.

• Proposed Solution: A production-ready architecture using an optimized XGBoost Regressor pipeline combined with trigonometric features. Raw timestamps are converted into 2D circular wave topologies (using Sine and Cosine conversions) to perfectly map adjacent intervals.

Code submission link:

https://github.com/shobhitatiwari22/upskillcampus/tree/main/Smart_City_Traffic_Management

Report submission link:

upskillcampus/Smart_City_Traffic_Management/SmartCityTrafficPatternDetection_Shobhita Tiwari_USC_UCT.pdf at main · shobhitatiwari22/upskillcampus

Proposed Design/ Model

The data lifecycle flows from multi-junction historical CSV logs through visual EDA notebooks to a serialized payload. The application interface allows operators to select a future date and junction, dynamically generating an absolute 24-hour sequence forecast.

High Level Diagram

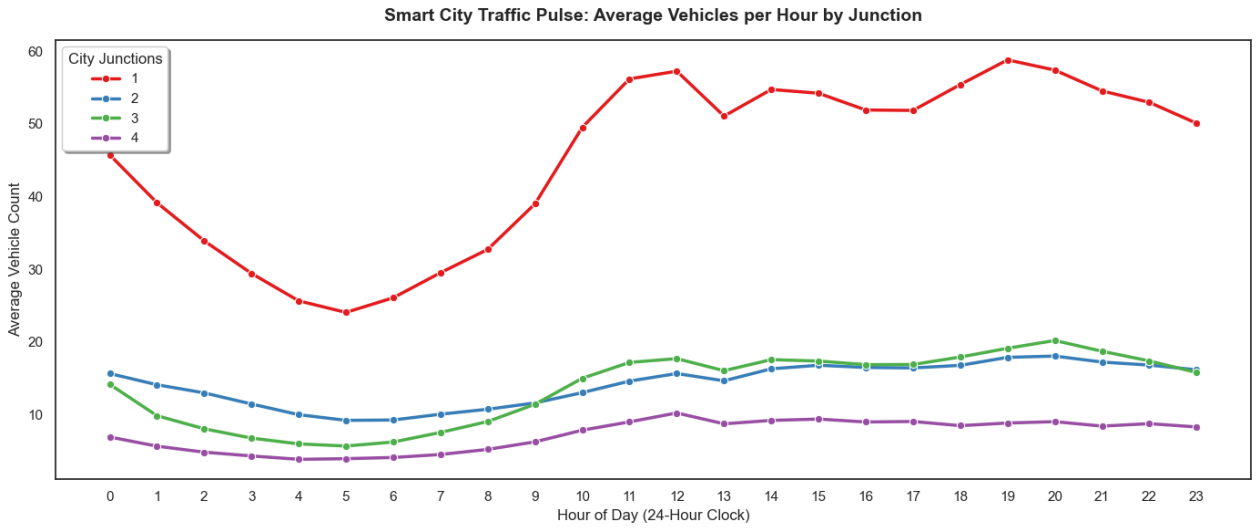
Historical Grid Data] ➔ [Time-Series Split] ➔ [Trigonometric Encodings] ➔ [XGBoost Training Pipeline] ➔ [Serialized Weights (.pkl)] ➔ [Streamlit Dashboard UI]

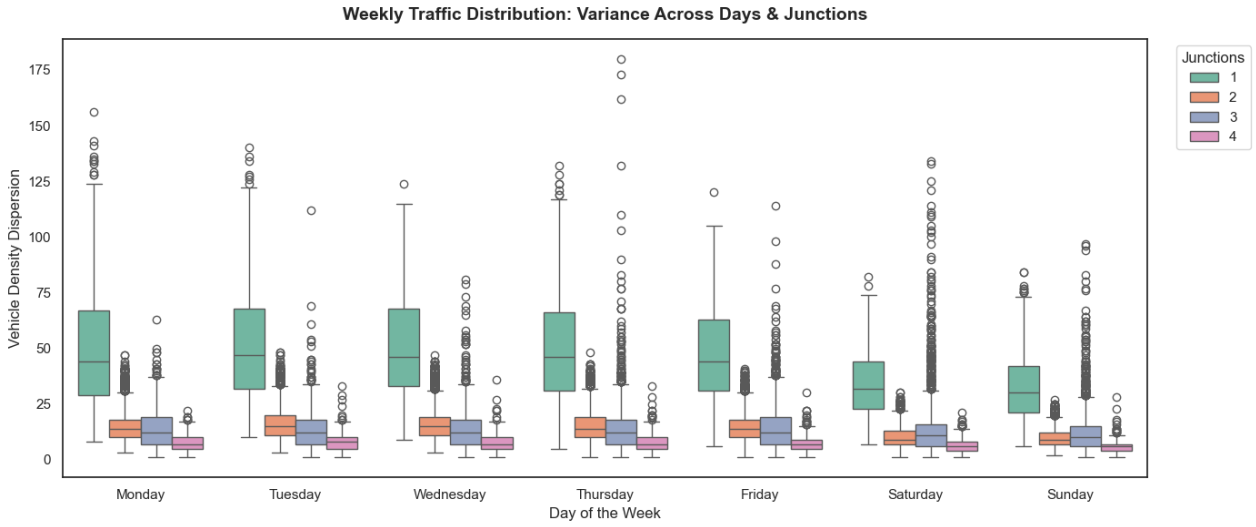
Low Level Diagram

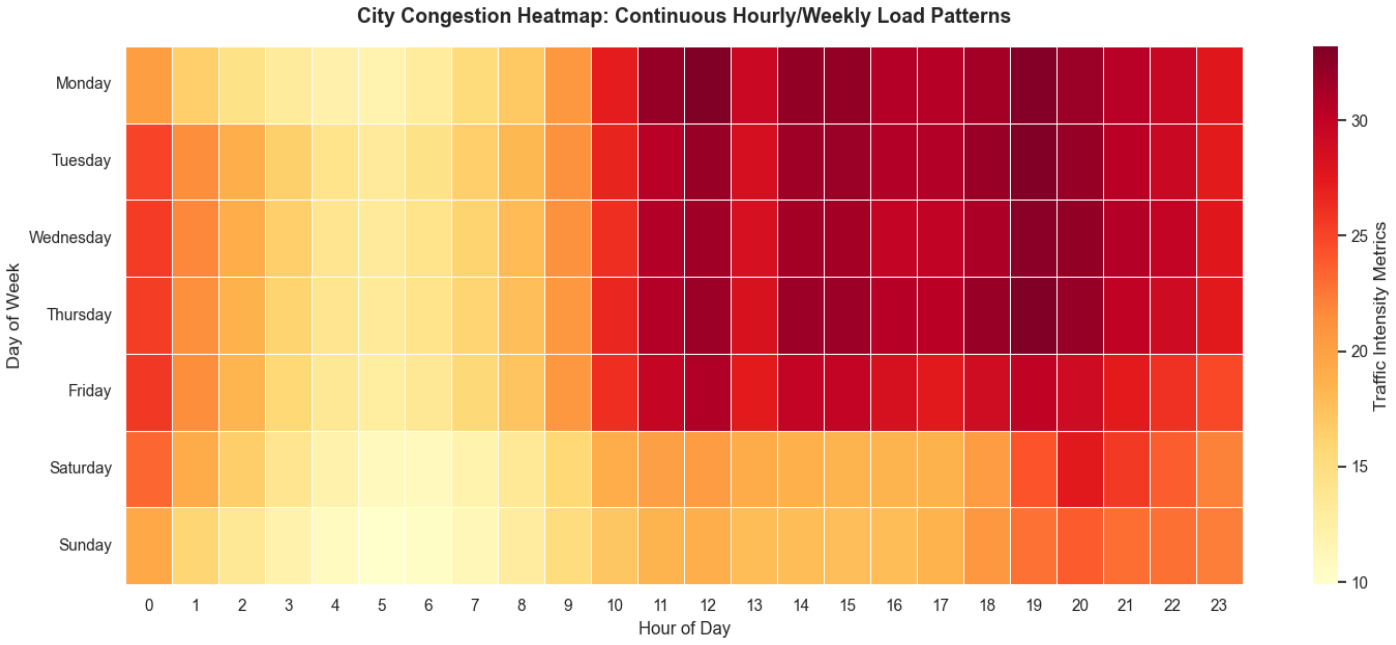
Raw Hour (0-23) ➔ Hour sin = sin(2*pi*Hour/24) ➔ [Predictive Arrays]

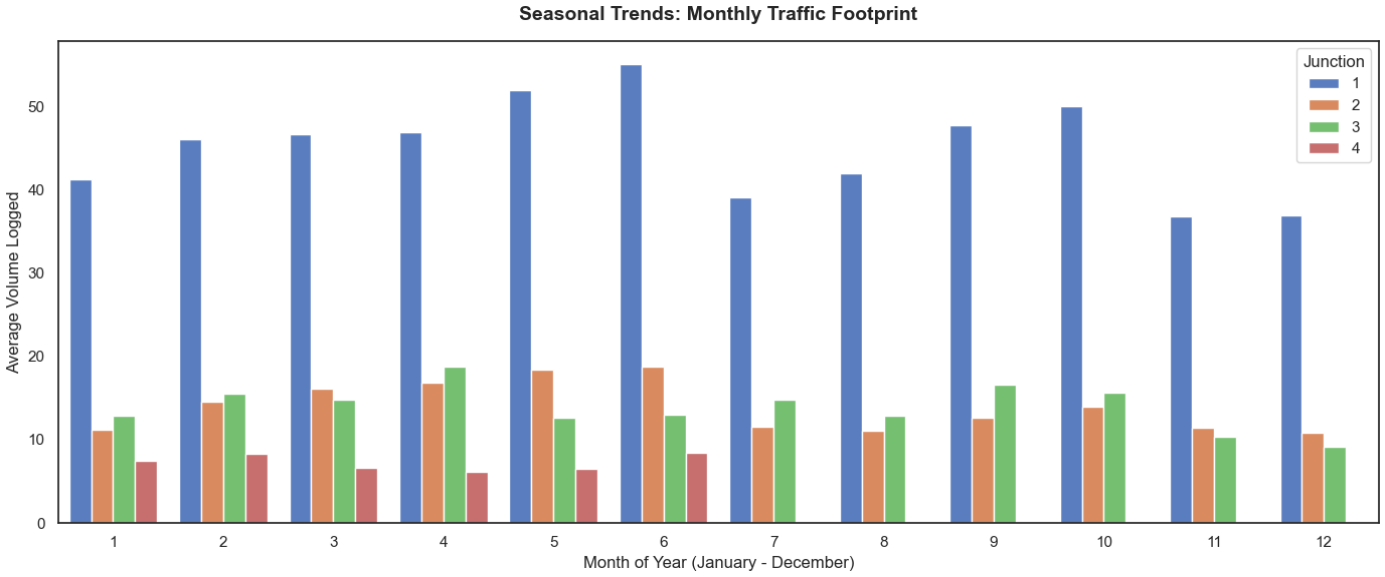
➔ Hour cos = cos(2*pi*Hour/24) ➔ [Target Variance Output]

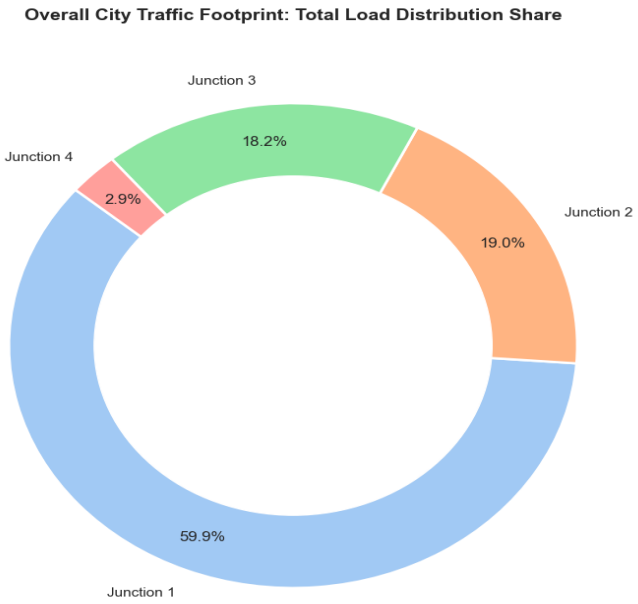
Interfaces

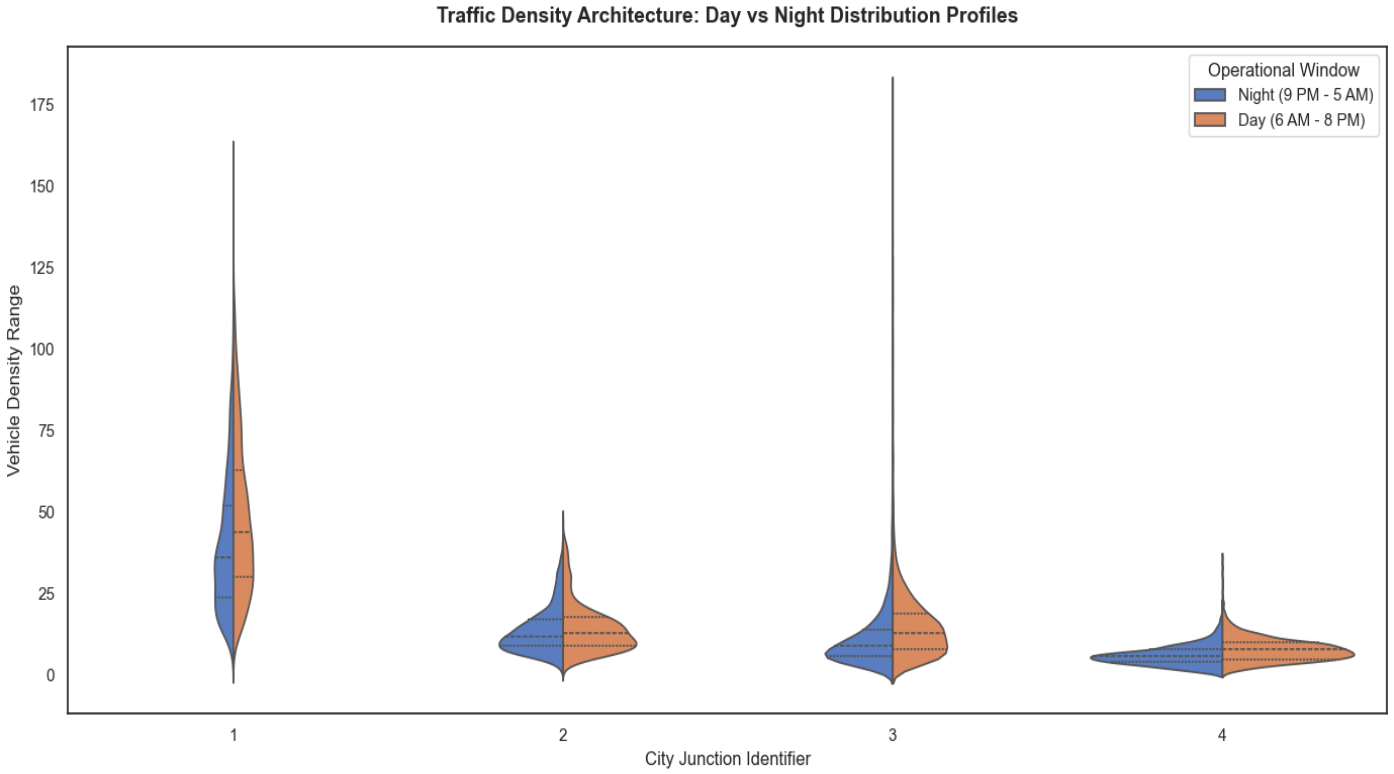

Smart City Traffic Pulse: Average Vehicles per Hour by Junction
City Junctions
1
2
3
4
Average Vehicle Count
Hour of Day (24-Hour Clock)


Weekly Traffic Distribution: Variance Across Days & Junctions
Junctions
1
2
3
4
Vehicle Density Dispersion
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Day of the Week


City Congestion Heatmap: Continuous Hourly/Weekly Load Patterns
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Day of Week
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Hour of Day
Traffic Intensity Metrics
10
15
20
25
30


Seasonal Trends: Monthly Traffic Footprint
Junction
1
2
3
4
Average Volume Logged
0
10
20
30
40
50
1 2 3 4 5 6 7 8 9 10 11 12
Month of Year (January - December)


Overall City Traffic Footprint: Total Load Distribution Share
Junction 3
18.2%
Junction 4
2.9%
Junction 2
19.0%
59.9%
Junction 1


Traffic Density Architecture: Day vs Night Distribution Profiles
Operational Window
Night (9 PM - 5 AM)
Day (6 AM - 8 PM)
Vehicle Density Range
175
150
125
100
75
50
25
0
1
2
3
4
City Junction Identifier

Performance Test

To ensure industrial stability, validation testing discarded random cross-validation in favor of a strict chronological sequence split, mirroring real-world conditions.

Test Plan / Test Cases

The system was tested against unseen future logging metrics across multiple locations, checking for model decay, system lag, and boundary extreme peaks.

Test Procedure

1. Load unseen evaluation datasets.

2. Parse time metrics through cyclical wave transformers.

3. Feed features into the serialized XGBoost engine(**traffic_production_model.pkl**) to generate predictive variations.

Performance Outcome

The system demonstrated elite forecasting accuracy on unseen validation data:

• Variance Explained (R^2 Score): 0.97 (97% accuracy without data leakage)

• Mean Absolute Error (MAE): 2.91 vehicles

• Root Mean Squared Error (RMSE): 4.78 vehicles

• Key Finding: Junction 1 was identified as the primary logistics hub, accounting for roughly 60% of total vehicular density with clear rush hour spikes at 9:00 AM and 6:00 PM.

My learnings

This project deepened my knowledge of advanced gradient-boosted decision tree arrays and spatial-temporal wave mapping. I learned how to build leakage-proof verification models, use Streamlit for cloud engine interfaces, and address industrial constraints like latency and data drift.

Future Work Scope

Future updates could integrate live streaming data layers using Kafka pipelines, connect deep learning networks like LSTMs for weather variables, and integrate the system with UCT Insight's dashboard via automated MQTT/HTTP protocols.